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RESEARCH ARTICLE

Diversity of clades I and II within *Anopheles funestus* sensu stricto in southern Africa

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Abstract *Anopheles funestus* sensu stricto is a major malaria vector in Africa. Recently, two clades were identified in this species based on mitochondrial DNA sequence data. We investigated differences between the two *An. funestus* clades based on nuclear genes and internal transcribed spacer 2 (ITS2) in the ribosomal DNA. Samples were collected from Malawi and Mozambique and were then used to sequence five nuclear genes (*BU21*, *9K1*, *6Z3, BU021* and *BU92*) and the ITS2 region. The DNA sequence data was used to assess the diversity of the two *An. funestus* clades. We identified 12, 10, 24, 10, and 21 polymorphisms in the two *An. funestus* clades in the *BU21, 9K1, 6Z3, BU021,* and *BU92* genes, respectively. No differences or polymorphisms were found in the ITS2 region between the two clades. This report assesses the diversity of the two *An. funestus* clades based on nuclear genes and the ITS2 region. Single nucleotide polymorphisms were found between the two clades of *An. funestus*. This could be explained by the existence of remaining introgression regions in the genomes of the two *An. funestus* clades and limited divergent regions. Further investigation is needed to determine the specific status of the two clades within *An. funestus*.

Keywords *Anopheles funestus* · Clade · Diversity · ITS2 · Nuclear gene · Single nucleotide polymorphisms (SNPs)

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Introduction

The *Anopheles funestus* group contains 11 morphologically similar adult African species (Gillies and Coetzee 1987; Harbach 2004) including a major malaria vector, *Anopheles funestus* sensu stricto, which is the most endophilic and anthropophilic member of the group (Gillies and De Meillon 1968). The other members of the group are mainly zoophilic and do not transmit malaria, although *Anopheles rivulorum* plays a minor role as a malaria vector in Tanzania (Wilkes et al. 1996), and *Anopheles vaneedeni* can transmit malaria under laboratory conditions (De Meillon et al. 1977). Recently, *An. funestus* was described with two cryptic subdivisions based on mitochondrial DNA (mtDNA) data, clade I which occurs throughout Africa, and clade II which is found only in Mozambique and Madagascar (Michel et al. 2005; Choi et al. 2012), and in Zambia (Choi et al. 2014). Furthermore, Michel et al. (2005) reported that there was an average 2 % divergence between the two clades, suggesting that they evolved independently around one million years ago. However, analysis of nuclear divergence using microsatellite loci in this same study did not support the mtDNA data.

Markers based on single nucleotide polymorphisms (SNPs) have been used for genetic mapping in eukaryotes to investigate population genetics, characterize candidate disease genes, and to study evolutionary processes (Morlais and Severson 2003). SNPs are the most common sequence variation in genomes, but they have only recently been used to examine the evolutionary history of populations, speciation, and historical demography (reviewed in Brumfield et al. 2003). A useful review of the application of SNPs in population genetics is provided by Morin et al. (2004). They concluded that SNPs might rapidly become the marker of choice for many applications in population

Table 1 Summary of specimens for DNA sequences

Clade	Country	Location	Sample ID	Access No.					
				BU21	9K1	6Z3	BU021	BU92	ITS2
I	Malawi	Nkhota kota (12°55′S, 34°18′E)	Mala43	KM261556	KM261537	KM261533	KM261547	KM261566	KM261571
			Mala44	KM261557	KM261538	KM261534			KM261572
		Karonga (10°19′S, 34°08′E)	Mala161	KM261558	KM261539	KM261535	KM261548	KM261567	KM261573
	Mozambique	Chibuto (24°40′S, 33°33′E)	MOZF25	KM261559	KM261543	KM261529	KM261551	KM261569	KM261577
			MOZF48		KM261544	KM261526	KM261553	KM261568	KM261578
			MOZF62		KM261546	KM261532		KM261570	KM261581
II	Mozambique	Chibuto (24°40′S, 33°33′E)	MOZF21		KM261540	KM261527	KM261555	KM261563	KM261574
			MOZF23		KM261541	KM261525	KM261549	KM261562	KM261575
			MOZF24		KM261542	KM261528	KM261550	KM261564	KM261576
			MOZF53	KM261561	KM261545	KM261530	KM261552	KM261565	KM261579
			MOZF58	KM261560	KM261536	KM261531	KM261554		KM261580

ecology, evolution, and conservation genetics, because of the potential for higher genotyping efficiency, data quality, genome-wide coverage, and analytical simplicity (e.g., in modeling mutational dynamics).

Here, we selected five nuclear genes which have high numbers of SNPs from Wondji et al. (2007) and the internal transcribed spacer 2 (ITS2) region in ribosomal DNA (rDNA) which may differ markedly between closely related species (Beckingham 1982; Porter and Collins 1991), and examined polymorphisms and nucleotide diversity in coding and non-coding regions of the five nuclear genes and ITS2 region from *An. funestus s.s.*

Materials and methods

Mosquito collection and identification

Mosquitoes were collected from houses in Malawi in 2001 and in Mozambique in 2004 using a sucking tube. Information regarding the specimens is given in Table 1. The identification of specimens was carried out according to the methods described by Gillies and Coetzee (1987) for morphology, and Koekemoer et al. (2002) for molecular diagnostics. The identification for two clades of *An. funestus* was performed using the method of Choi et al. (2013).

Laboratory methods and data analysis

All DNA samples were extracted from either single mosquitoes or available parts of mosquitoes using standard protocols (Ballinger-Crabtree et al. 1992) before being resuspended in Tris–EDTA (TE) buffer at volumes between 50 and 100 µL. The regions of interest were amplified using six pairs of primers (Table 2). Amplification of five nuclear gene regions was performed in a total volume of 50 µL containing 1 µL of genomic DNA from each sample, 1 × PCR Buffer, 2 mM $MgCl_2$, 0.2 mM of each dNTP, 0.2 µM of each primer, and 1 unit of Taq DNA polymerase. PCR cycling conditions for the five regions were as follows: a 5 min 94 °C denaturation step followed by 30–35 cycles of 30 s at 94 °C, 30 s at 55–57 °C, and 30 s at 72 °C; there was a final extension step of 10 min at 72 °C. Amplification of the ITS2 region was performed using the method of Koekemoer et al. (2002). All samples

Table 2 Summary of PCR primers used

Primer name	Primer sequences 5′-3′	Published by
BU21		Wondji et al. (2007)
Forward	TTT CAA GGT GAA CGG TGT GA	
Reverse	CCA TCA AGA TGA CGA CCA GA	
9K1		Wondji et al. (2007)
Forward	GTA CGA GCT GGC CGT TAA TC	
Reverse	CCT TTC TGT AGC TGC ACC TTG	
6Z3		Wondji et al. (2007)
Forward	GAC GAT CCG TTC CTG AAG AC	
Reverse	ATC GGT AAG CCC GGA TAT TT	
BU021		Wondji et al. (2007)
Forward	GAG TTG GTT GAT GCC GTG TT	
Reverse	CGT CCG GAA ACA AAT ATC GT	
BU92		Wondji et al. (2007)
Forward	CAT GCG ACC GAA GAG AAG TT	
Reverse	ATC CTG ATT CTG GCT CAT GG	
ITS2		Koekemoer et al. (2002)
Forward	TGT GAA CTG CAG GAC ACA T	
Reverse	TAT GCT TAA ATT CAG GGG GT	

Table 3 Information of single nucleotide polymorphisms (SNPs)

Sequence region	Chromosomal location (Division)	Function	Product length	No. of SNPs
BU21	X (3A)	Phosphoribosylaminoimida-zole carboxylase	456 bp (25 + 91[N] + 340)	12 (6 + 6[N])
9K1	X (5)	Cytochrome P450	243 bp	10
6Z3	2L (25A-27D)	Cytochrome P450	515 bp	24
BU021	3L (38C-40B)	Tubilin beta-3 chain	377 bp (64 + 65[N] + 248)	10 (5 + 5[N])
BU92	3L (39A)	Microtubule binding	499 bp (89 + 112[N] + 298)	21 (8 + 13[N])
ITS2			488 bp	0

[N] represents non-coding regions

were sequenced at Macrogen Inc. in Korea using an ABI 3730XL DNA analyzer (Applied Biosystems, Foster City, CA, USA).

The DNA sequence data were aligned using the ClustalW algorithm in Bioedit 7.2.5 (Hall 1999) and the sequences were deposited in GenBank under accession numbers (KM261525-KM261581) (Table 1). Sequence polymorphism (*Pi*) was estimated using DnaSP 5.0 (Rozas et al. 2003).

Results and discussion

Table 3 presents the number of SNPs and the length of sequenced regions of the five studied genes and ITS2. *BU21*, *BU021*, and *BU92* have 6, 5, and 8 SNPs in coding regions and 6, 5, and 13 SNPs in non-coding regions, respectively. The remaining two genes, *9K1* and *6Z3* have 10 and 24 SNPs in coding regions, respectively. The number of SNPs in this study were 1–4 SNPs more than those of Wondji et al. (2007). It might be due to local specificity of the samples. There was no differentiation between the two *An. funestus* clades in the studied regions. Furthermore, we found no differences or SNPs between the two clades in the ITS2 region. A large number of SNPs were located in the beginning of the sequenced regions except the 6Z3 gene. Figure 1 shows the peaks of SNPs in the five studied nuclear genes.

Garros et al. (2004) suggested that there are two genotypes of *An. funestus*, labeled M and W, identified by Domain III in rDNA using restriction fragment length polymorphism (RFLP) typing. Type M is found in East Africa and Malagasy, while type W occurs in West Africa. The samples from South Africa contained both M and W (MW) types. Koekemoer et al. (2006) reported that there were two additional RFLP types; one found in Malawi (Y type) and one in Angola, Malawi, Ghana, and Zambia (Z type). Furthermore, Coetzee and Koekemoer (2013) reviewed the molecular systematics of *An. funestus* which may indicate simple geographic variation, although two distinct clades from the analyses of mtDNA in southern Africa are seemed to reflect that a speciation event has happened. However, there were no differences between the two clades of *An. funestus* from the analysis of the sporozoite infection rates and insecticide resistance (Choi et al. 2014).

Single nucleotide polymorphisms occur at a rate of around one per kilobase of DNA in humans (Wang et al. 1998) and one per 125 base pairs in *Anopheles gambiae* (Morlais and Severson 2003). Wondji et al. (2007) screened 20,547 bp from 50 genes of *An. funestus* and identified 303 SNPs in 14,671 bp of coding regions, and 191 SNPs in 5,876 bp of non-coding regions. This study concluded that SNPs in *An. funestus* had been influenced by recombination, the degeneracy of the genetic code, and selective constraints of gene conservation. Furthermore, they concluded that SNPs will be important in further studies on population genetics, for monitoring insecticide resistance genes, and in the investigation of genes related to vector competence.

Although no fixed differences were found between the two clades of *An. funestus* in this study, we found that various SNPs exist in nuclear genes within clades of this species. Choi et al. (2012) indicated that the divergence between two clades of *An. funestus* may be limited only to mtDNA although distinct subdivisions in the species exist. Michel et al. (2005) also reported that there was deep mtDNA divergence between two clades of *An. funestus* but not found any corresponding nuclear divergence. They suggested that historical introgression either with a closely related species or among previously separated and divergent populations may result in the subdivisions of the species (Michel et al. 2005). Furthermore, the nature of mtDNA such as non-recombination, maternal heritage and haploid could retain both apparent mtDNA subdivisions in admixed populations, while the nuclear regions in the species would become homogenized. In a review on *An. funestus*, Coetzee and Koekemoer (2013) commented that another major malaria vector, *An. gambiae* showed a similar pattern to the M and S molecular forms (designated

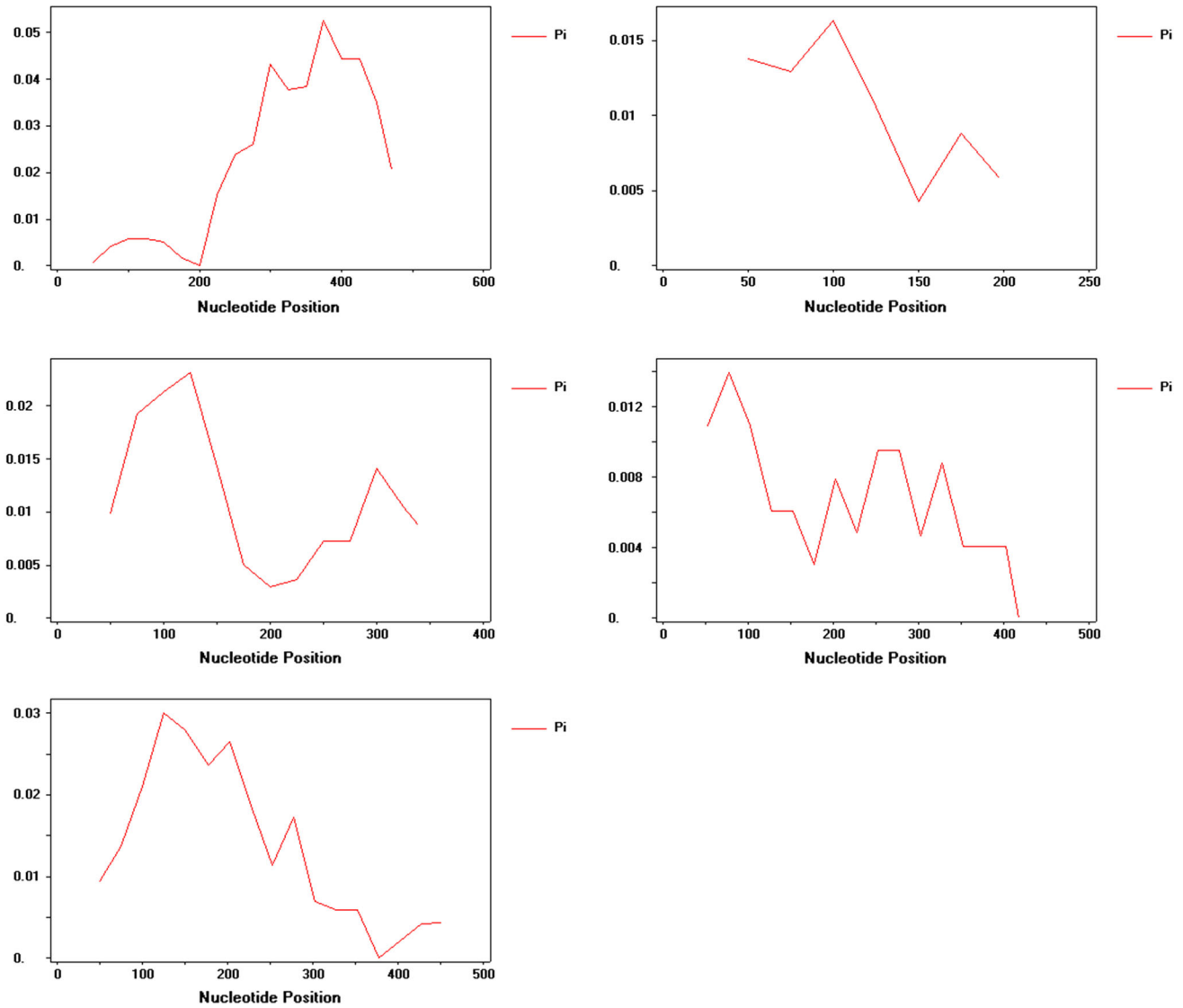


Fig. 1 Polymorphisms in five nuclear genes from *An. funestus*

Anopheles coluzzii and *An. gambiae* respectively in Coetzee et al. 2013). However, this diversity within *An. funestus* remains unclear and further studies are needed to increase sample sizes in the study regions and other genomic regions, and subsequently to determine what the status of these clades means in terms of species differentiation.

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Conflict of interest The authors declare no conflict of interest.

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